

Software Requirement Specification

Objective

This SRS document will explain the functional and non-functional requirements for an online problem-solving website. The platform is basically designed to support coding practice, competitive programming, and interview preparation and tutorials. The document is created to support developers and testers to meet the user and business requirement.

Important features:

The website will allow users to solve problems, participate in contests, and track progress.

It Include community support and collaboration features.

Provide admin and judge tools for problem, contest, and user management.

Implement gamification features for participant on contest such as awards, badges, and leaderboards.

Target Audience: Who are school and college students intended to develop programming skill.

Timeline: The system is expected to be fully functional within 2 months.

Accessibility:

Admin: User with full control over the system.

Judge: User responsible for evaluating solutions and managing test cases.

Users: user should be authorized by registration method

2. Overall Description

Product Perspective

The system will be a web-based application accessible from desktops, tablets, and mobile devices. It will integrate a database for storing user profiles, problems, contests, and activity logs.

Website functionalities:

User registration and profile management

Problem submission, categorization, and display

Community support for discussions and collaboration

Contest creation and management

Admin and judge functionalities for monitoring and evaluation

Gamification with points, badges, and leaderboards

Search, filter, and discovery of problems based on the difficulty or topic tag

User Classes and Characteristics

Admin must have capability of management, security, monitoring and update.

Judge should have the access to set test cases, evaluate submissions, and manage problem data.

Only registered users should have the access to solve the problems, participate on contest and learning resources.

Non-registered can only explore the website.

Operating Environment:

Web browsers: Chrome, Firefox, Safari, Edge

Devices: Desktop, tablet, mobile

Front-end: HTML, Tailwind CSS, Vue.js, Javascript

Backend: PHP/Laravel

Database: MySQL

Design and Implementation Constraints

Development must be completed within 2 months

Secure handling of user data and authentication

The website should be capable of huge number of user pressure with minimal problems.

Dependencies

System may use third-party libraries for messaging, gamification, and notifications

3. Functional Requirements

User Management

Users must register and log in to access services

Each user's profile must reflect his/her interest and programming skill

Password recovery option is required

Guest users can view home page content only

Problem Submission and Display

Users can submit problems with categories, tags, and media (images/videos). So that it can be identified easily.

Users can comment and discuss problems.

Problem Set Management

Admin can create, update, or delete problems

Judges can update problem test cases

Problems are categorized by difficulty and topic

Search and Discovery

Users can filter problems by topic and difficulty

Weekly suggested problem lists displayed to encourage practice

Integration and Collaboration:

Users can comment, rate, or upvote posts

One-to-one messaging and group chat available for collaboration

Gamification and User Engagement:

Users earn points, badges, or awards for solving problems

Leaderboards display top contributors and top contest performers

4. Non-Functional Requirements

Performance

The routing and fetching of data should be fast and it should be responsive for all device.

Security

Secure login, authentication, and data protection against unauthorized access

Usability

Intuitive and easy to navigate for users of all technical levels

Compatibility

Supports all major browsers and devices (desktop, tablet, mobile)

Reliability

High uptime to ensure availability whenever users need access

5. System Architecture Overview

Frontend: Web interface for users, admins, and judges

Backend: Handles authentication, business logic, problem evaluation, and notifications

Database: Stores user data, problems, contests, and logs

APIs: For communication between frontend and backend

Progress of website

How many problems have been solved on the website by all users.

User engagement and active participation

Growth of community and contest participation